INCH-POUND

MIL-S-22820/7C
28 November 1990
SUPERSEDING
MIL-S-22820/7B
1 November 1972

MILITARY SPECIFICATION SHEET
SERVOMOTOR-TACHOMETER GENERATOR, TYPE 18SM-RG4D

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Servomotor-Tachometer Generators described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-S-22820.

TABLE I. Requirements of servomotor section.

Requirement	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	hz	
Test	400	hz	±1%
Range	400	hz	±5%
Voltage:			
Reference phase, rated	115	volts	
Reference phase, test	115	volts	±1%
Reference phase, range	115	volts	±10%
Control phase, rated (series)	36	volts	
Control phase, test (series)	36	volts	±1%
Control phase, range (series)	0 to 39.6	volts	
Control phase, rated (parallel)	18	volts	
Control phase, range (parallel)	0 to 19.8	volts	
Current:			
Reference phase	0.200	amperes	maximum
Control phase (series)	0.634	amperes	maximum
Control phase (parallel)	1.30	amperes	maximum
Power:			
Reference phase	9.1	watts	maximum
Control phase (series)	9.1	watts	maximum
Control phase (parallel)	9.1	watts	maximum
Synchronous speed	6,000	rpm	
Impedance:			
Reference phase			
Control phase (series)	266+j580	ohms	±30%
Control phase (parallel)	26+j57	ohms	±30%
Power factor:			
Reference phase	0.42		±0.10
Control phase	0.42		±0.10
Electrical breakaway torque	1.0	volts	maximum
Coupling in control winding	0.7	ratio	minimum

AMSC - N/A

FSC 6105

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TABLE I. Requirements of servomotor section - Continued.

Requirements	Value	Unit	Tolerance
Mechanical:			
No-load speed	4,500	rpm	minimum
Stall torque at 23°C	2.20	ounce-inches	minimum
Rotor moment of enertio	5.73	gm-cm^2	maximum
Radial play	0.0008	inches	maximum
End play	0.0010-0.0035	inches	min.-max.
Temperature rise	85.0	degrees C	maximum
Time constant	0.025	seconds	maximum
Reversing time	0.040	seconds	maximum
Maximum power output	1.1	watts	minimum
Weight	21.0	ounces	maximum

TABLE II. Requirements for tachometer-generator section.

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	Hz	
Test	400	Hz	±1%
Range	400	Hz	±5%
Voltage:			
Energizing winding, rated	115	volts	
Energizing winding, test	115	volts	±1%
Energizing winding, range	115	volts	±10%
Current:			
Energizing winding	0.62-0.78	amperes	min.-max.
Power:			
Energizing winding	5.6	watts	maximum
Power factor:			
Energizing winding	0.63	-----	±0.1
Impedance:			
Energizing winding	1,035+J1,276	ohms	±30%
Output winding	2,500-3,000	ohms	min.-max.
In-phase speed-sensitive transformation ratio (at 3000 rpm) 1/	0.0806	ratio	±2.5%
Phase shift (at 3000 rpm) 1/	4.5 (lag)	degrees	±1.5
In-phase linearity 1/	±0.8	percent	
In-phase axis error 1/	0.008	volts	maximum
Zero-speed output voltage 1/			
Total	0.013	volts	maximum
Fundamental	0.010	volts	maximum

1/Measured open circuit for values specified in Table II.

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TABLE III. Part identifying number variant characteristics

Part Identifying Number 1/	Shaft Style	Termination	G
M22820/7-01A	Splined	Terminal Board	3.281

1/Part identifying number M22820/7-01A shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE IV. Part identifying number cross references.

SUPERSEDED PART NUMBER	NEW PART IDENTIFYING NUMBER
BUORD Drawing 2009026	
MK 16 Mod 3	M22820/7-01A

1/All the line items shown in Table IV refer to equivalent and interchangeable servomotors of the same type designation modification. Part number changes do not affect form, fit, or function of the servomotor-tachometer generators listed therein.

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NOTES:

1. Type 18SM-RG4D servomotor-tachometer generators shall be capable of operation in ambient temperatures ranging from -55°C to 100°C.

2. The speed range for the 18SM-RG4D is 0 to 3,600 rpm. The calibration speed for linearity requirements shall be 3,000 rpm. Phase shift and transformation ratio tests shall be conducted at calibration speed.

3. The servomotor-tachometer generator identified and specified by this specification shall conform to all requirements and shall be subjected to all tests delineated in MIL-S-22820, with the following exceptions:

   a. Voltage sensitivity (tachometer generator section)
   b. Frequency sensitivity (tachometer generator section)
   c. Temperature sensitivity (tachometer generator section)
   d. Warm-up time
   e. Position-error voltage

4. If, during measurement, the ambient temperature is in the range of temperatures defined by 23° ± 5°C, the corrected transformation ratio shall be:

$$TR_C = \frac{TR_O}{1 - \frac{(t_O - 23)}{400}}$$

where: TR_C = Corrected transformation ratio
TR_O = Measured transformation ratio
t_O = Ambient temperature at which TR_O was measured

The corrected transformation ratio shall be within the limits specified in Table II herein. Similarly, the corrected lagging phase shift angle shall be:

$$\phi_C \text{ degrees} = \phi_O \text{ degrees} - 1/8\ (t_O - 23) \text{ degrees}$$

where: ϕ_C degrees = Corrected lagging phase shift angle
ϕ_O degrees = Measured lagging phase shift angle
t_O = Ambient temperature at which ϕ was measured

Custodian:
Army - AR
Navy - AS
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Preparing Activity:
Navy - AS

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Review Activities:
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